

KWEKU AGYEMANG

Lecturer in Computer Science

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PROFESSIONAL SUMMARY

A passionate and research-driven Computer Science professional with over 8 years of combined experience in academia and industry. Committed to delivering innovative, student-centred teaching in algorithms, software engineering, and artificial intelligence. Seeks to contribute to the academic mission of Pentecost University by nurturing the next generation of technology leaders and fostering a culture of research excellence.

EDUCATION

Doctor of Philosophy (PhD) - Computer Science

2016 - 2020

University of Ghana, Legon, Accra

- Thesis: Adaptive Machine Learning Frameworks for Resource-Constrained Environments in Sub-Saharan Africa
- Supervised by Prof. Kofi Mensah; graduated with distinction

Master of Science - Computer Science

2013 - 2015

Kwame Nkrumah University of Science and Technology (KNUST), Kumasi

- Specialisation in Software Engineering and Distributed Systems
- Graduated with First Class Honours; Best Graduating Student Award

Bachelor of Science - Computer Science

2009 - 2013

University of Cape Coast, Cape Coast

- Dean's List, 2011, 2012, 2013
- Final Year Project: Mobile-Based Health Record Management System for Rural Clinics

ACADEMIC EXPERIENCE

Assistant Lecturer - Computer Science

Jan 2021 - Present

Ghana Communication Technology University (GCTU), Accra

- Deliver lectures and tutorials in Data Structures & Algorithms, Object-Oriented Programming, and Database Systems to cohorts of 60-120 students
- Develop and update course materials, syllabi, and assessments in line with university curriculum standards
- Supervise final year undergraduate projects; mentored 18 students to successful completion
- Coordinate departmental seminar series on emerging technologies in AI and cybersecurity
- Achieved average student satisfaction rating of 4.7/5.0 over four academic years

Graduate Teaching Assistant

2016 - 2020

University of Ghana, Legon

- Conducted tutorials and lab sessions in Computer Networks, Operating Systems, and Introduction to Programming (Python)
- Assisted in grading, mentoring, and preparing course materials for undergraduate modules

RESEARCH & PUBLICATIONS

Research Interests: Machine Learning, AI Ethics, Mobile Computing, Software Engineering, ICT4D

Selected Peer-Reviewed Publications:

- Agyemang, K., & Mensah, K. (2022). Lightweight Neural Network Architectures for Mobile Platforms in Low-Bandwidth Environments. *Journal of African Research in Computing*, 14(2), 45-61.
- Agyemang, K. (2021). Towards Ethical AI Deployment in Healthcare Systems in Ghana. *Proceedings of the 3rd African Conference on Artificial Intelligence (ACAI 2021)*, Nairobi.
- Agyemang, K., Boateng, R., & Asante, E. (2020). A Framework for Adaptive Data Compression in Resource-Constrained IoT Devices. *International Journal of Computer Science and Engineering*, 9(4), 112-128.
- Agyemang, K. (2019). Mobile-First Database Solutions for Healthcare Record Management in Rural Ghana. *African Journal of Information Systems*, 11(3), 77-94.

INDUSTRY EXPERIENCE

Software Engineer (Consultancy)

Rancard Solutions, Accra, Ghana

2015 - 2016

- Developed back-end services using Java and Python for mobile data platforms serving over 500,000 users
- Collaborated in agile teams to deliver features for SMS-based financial services applications

IT Systems Intern

Ghana Revenue Authority, Accra, Ghana

2012 - 2013

- Assisted in database administration and system maintenance for internal reporting tools
- Developed a VBA-based reporting tool that reduced weekly report preparation time by 40%

TECHNICAL SKILLS

Programming Languages	Python, Java, C++, JavaScript, SQL, R
Frameworks & Tools	TensorFlow, PyTorch, Django, React, Git, Docker, MATLAB
Teaching Tools	Moodle LMS, Google Classroom, MS Teams, Kahoot, Turnitin
Research Methods	Quantitative & Qualitative Analysis, Systematic Literature Review, Survey Design
Languages	English (Fluent), Twi (Native), French (Intermediate)

PROFESSIONAL DEVELOPMENT & CERTIFICATIONS

- Google Certificate in Machine Learning (Coursera) - 2022
- Higher Education Teaching Certificate - National Teaching Council, Ghana - 2021
- AWS Certified Cloud Practitioner - 2020
- Participant, Pan-African Higher Education Teaching Excellence Programme - DAAD, 2019

SERVICE & PROFESSIONAL AFFILIATIONS

- Member, Ghana Institution of Engineering (GhIE)
- Member, Association for Computing Machinery (ACM)
- Reviewer, African Journal of Information Systems (2021 - Present)
- Departmental Examinations Officer, GCTU (2022 - Present)
- Volunteer Mentor, Girls in ICT Ghana Initiative (2019 - Present)

REFEREES

Available upon request.